№2

:

2026-03-19

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1. 1.
2. 2.
3. 3.

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1.

1.1

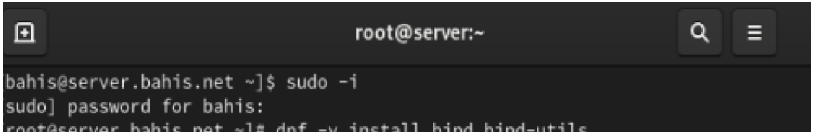
-
-
- Markdown

2

2.

2.1 DNS-

- bahis
- sudo -i
- root



1: sudo -i

- bind bind-utils
- dnf -y install
-

```
[root@server.bahis.net ~]# dnf -y install bind bind-utils
Extra Packages for Enterprise Linux 9 - x86_64   43 kB/s |  36 kB     00:00
Extra Packages for Enterprise Linux 9 - x86_64  6.8 MB/s |  20 MB     00:02
Rocky Linux 9 - BaseOS                          8.3 kB/s | 4.3 kB     00:00
Rocky Linux 9 - AppStream                        15 kB/s | 4.8 kB     00:00
Rocky Linux 9 - Extras                          9.6 kB/s | 3.1 kB     00:00
Package bind-utils-32:9.16.23-34.el9_7.1.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package                Arch       Version                    Repository     Size
================================================================================
Installing:
 bind                   x86_64     32:9.16.23-34.el9_7.1      appstream     488 k
Installing dependencies:
 bind-dnssec-doc        noarch     32:9.16.23-34.el9_7.1      appstream      45 k
 python3-bind           noarch     32:9.16.23-34.el9_7.1      appstream      61 k
 python3-ply            noarch     3.11-14.el9.0.1            baseos        103 k
Installing weak dependencies:
 bind-dnssec-utils      x86_64     32:9.16.23-34.el9_7.1      appstream     113 k

Transaction Summary
```

2: bind bind-utils dnf

- dig www.yandex.ru
- NOERROR
- A-
-

```
root@server.bahis.net ~]# dig www.yandex.ru

 <<>> DiG 9.16.23-RH <<>> www.yandex.ru
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3356
; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

; OPT PSEUDOSECTION:
 EDNS: version: 0, flags:; MBZ: 0x001f, udp: 512
; QUESTION SECTION:
www.yandex.ru.                  IN      A

; ANSWER SECTION:
ww.yandex.ru.           31      IN      A       77.88.55.88
ww.yandex.ru.           31      IN      A       5.255.255.77
ww.yandex.ru.           31      IN      A       77.88.44.55

; Query time: 23 msec
; SERVER: 172.9.0.14#53(172.9.0.14)
; WHEN: Fri Feb 06 18:16:19 UTC 2026
```

- /etc/resolv.conf
- bahis.net
- DNS-

```
[root@server.bahis.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search bahis.net
nameserver 172.9.0.14
nameserver 192.168.150.36
[root@server.bahis.net ~]#
```

4: /etc/resolv.conf

- /etc/named.conf
- recursion yes
- 127.0.0.1
- localhost

```
[root@server.bahis.net ~]# cat /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
        listen-on port 53 { 127.0.0.1; };
        listen-on-v6 port 53 { ::1; };
        directory       "/var/named";
        dump-file       "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
        memstatistics-file "/var/named/data/named_mem_stats.txt";
        secroots-file   "/var/named/data/named.secroots";
        recursing-file  "/var/named/data/named.recursing";
```

- `named.ca`
- NS, A, AAAA-

```
[root@server.bahis.net ~]# cat /var/named/named.ca

; <<>> DiG 9.18.20 <<>> -4 +tcp +norec +nostats @d.root-servers.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47286
;; flags: qr aa; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 27

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1450
;; QUESTION SECTION:
;.                              IN      NS

;; ANSWER SECTION:
.                       518400  IN      NS      a.root-servers.net.
.                       518400  IN      NS      b.root-servers.net.
.                       518400  IN      NS      c.root-servers.net.
.                       518400  IN      NS      d.root-servers.net.
.                       518400  IN      NS      e.root-servers.net.
.                       518400  IN      NS      f.root-servers.net.
```

- named.localhost
- localhost

```
[root@server.bahis.net ~]# cat /var/named/named.localhost
$TTL 1D
@       IN SOA  @ rname.invalid. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        NS      @
        A       127.0.0.1
        AAAA    ::1
[root@server.bahis.net ~]#
```

7: named.localhost

- named.loopback
- 127.0.0.1

```
[root@server.bahis.net ~]# cat /var/named/named.loopback
$TTL 1D
@       IN SOA  @ rname.invalid. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        NS      @
        A       127.0.0.1
        AAAA    ::1
        PTR     localhost.
[root@server.bahis.net ~]#
```

8: named.loopback

- `systemctl start named  enable named`
- `dig`
-

```
[root@server.bahis.net ~]# systemctl start named
[root@server.bahis.net ~]# systemctl enable named
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/syste
m/named.service.
[root@server.bahis.net ~]# dig www.yandex.ru

; <<>> DiG 9.16.23-RH <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 53892
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;www.yandex.ru.                 IN      A

;; ANSWER SECTION:
www.yandex.ru.          201     IN      A       77.88.44.55
www.yandex.ru.          201     IN      A       77.88.55.88
www.yandex.ru.          201     IN      A       5.255.255.77

;; Query time: 5 msec
;; SERVER: 172.9.0.14#53(172.9.0.14)
;; WHEN: Fri Feb 06 18:36:30 UTC 2026
;; MSG SIZE  rcvd: 118

[root@server.bahis.net ~]# dig @127.0.0.1 www.yandex.ru
```

- `nmcli` DNS 127.0.0.1
- auto-DNS
- `/etc/resolv.conf`

```
[root@server.bahis.net ~]# nmcli connection edit eth0
remove ipv4.dns

===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-1x, dcb, sriov, et
htool, match, ipv4, ipv6, prefix-delegation, hostname, link, tc, proxy
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
save
nmcli> save
Error: connection verification failed: ipv4.dns: 2. DNS server address is invalid
You may try running 'verify fix' to fix errors.
nmcli> quit
The connection is not saved. Do you really want to quit? (yes/no) [no] yes
bash: remove: command not found...
[root@server.bahis.net ~]# nmcli connection edit System\ eth0

===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'System eth0'
```

- named.conf listen-on {127.0.0.1; any;}
- 192.168.0.0/16
- firewall DNS

```
[root@server.bahis.net ~]# cat /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
        listen-on port 53 { 127.0.0.1; };
        listen-on-v6 port 53 { ::1; };
        directory       "/var/named";
        dump-file       "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
        memstatistics-file "/var/named/data/named_mem_stats.txt";
        secroots-file   "/var/named/data/named.secroots";
        recursing-file  "/var/named/data/named.recursing";
        allow-query     { localhost; };

        /*
         - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
         - If you are building a RECURSIVE (caching) DNS server, you need to enable
           recursion.
         - If your recursive DNS server has a public IP address, you MUST enable access
           control to limit queries to your legitimate users. Failing to do so will
```

- 53
- named (UDP)

```
[sudo] password for bahis:
[root@server.bahis.net ~]# firewall-cmd --add-service=dns
Warning: ALREADY_ENABLED: 'dns' already in 'public'
success
[root@server.bahis.net ~]# firewall-cmd --add-service=dns --permanent
Warning: ALREADY_ENABLED: dns
success
[root@server.bahis.net ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
      Output information may be incomplete.
avahi-dae  590                          avahi   12u     IPv4              21423
     0t0        UDP *:mdns
avahi-dae  590                          avahi   13u     IPv6              21424
     0t0        UDP *:mdns
chronyd    633                         chrony    5u     IPv4              21467
     0t0        UDP localhost:323
chronyd    633                         chrony    6u     IPv6              21468
     0t0        UDP localhost:323
named      974                          named   21u     IPv4              23041
     0t0        UDP localhost:domain
```

- forwarders {127.0.0.1;}
- forward first
- dnssec-enable dnssec-validation

root@server:/etc/kea

```
            // This governs whether the log output is flushed to disk after
            // every write.
            // "flush": false,

            // This specifies the maximum size of the file before it is
            // rotated.
            // "maxsize": 1048576,

            // This specifies the maximum number of rotated files to keep.
            // "maxver": 8
        }
    ],
    // This specifies the severity of log messages to keep. Supported values
    // are: FATAL, ERROR, WARN, INFO, DEBUG
    "severity": "INFO",

    // If DEBUG level is specified, this value is used. 0 is least verbose,
    // 99 is most verbose. Be cautious, Kea can generate lots and lots
    // of logs if told to do so.
    "debuglevel": 0
  }
]
```

- named.rfc1912.zones
- bahis.net
- include named.conf

```
[root@server.bahis.net ~]# cp /etc/named.rfc1912.zones /etc/named/
[root@server.bahis.net ~]# cd /etc/named
[root@server.bahis.net named]# mv /etc/named/named.rfc1912.zones /etc/named/user
.net
[root@server.bahis.net named]# vim /etc/named.conf
```

14: bahis.net named.conf

- bahis.net
- 1.168.192.in-addr.arpa
-

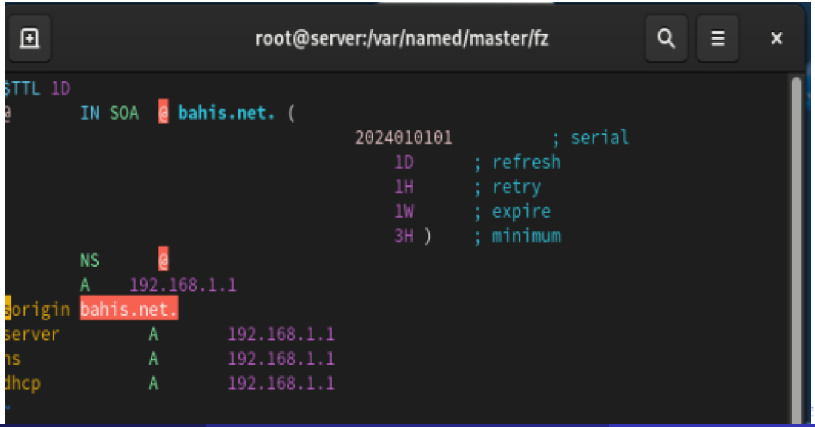
```
        };
};

zone "." IN {
        type hint;
        file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key
qa
```

- `master/fz  master/rz`
-

```
[root@server.bahis.net ~]# cd /var/named
[root@server.bahis.net named]# mkdir -p /var/named/master/fz
[root@server.bahis.net named]# mkdir -p /var/named/master/rz
[root@server.bahis.net named]# cp /var/named/named.localhost /var/named/master/f
z/
[root@server.bahis.net named]# cd /var/named/master/fz/
[root@server.bahis.net fz]# mv named.localhost bahis.net
[root@server.bahis.net fz]# ls
bahis.net
[root@server.bahis.net fz]#
```

16: master/fz master/rz

-
- $TTL 1D
- SOA-
-
- A- PTR-
- $ORIGIN



- SELinux-
- named_write_master_zones
- systemctl restart named

```
[root@server.bahis.net rz]# restorecon -vR /etc
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to un
confined_u:object_r:net_conf_t:s0
Relabeled /etc/cups/cupsd.conf.rpmsave from unconfined_u:object_r:cupsd_etc_t:s0 to unconfined_u:o
bject_r:cupsd_rw_etc_t:s0
[root@server.bahis.net rz]# restorecon -vR /var/named
[root@server.bahis.net rz]# getsebool -a | grep named
named_tcp_bind_http_port --> off
named_write_master_zones --> on
[root@server.bahis.net rz]# setsebool named_write_master_zones 1
[root@server.bahis.net rz]# setsebool -P named_write_master_zones 1
[root@server.bahis.net rz]# journalctl -x -f
Feb 07 13:41:27 server.bahis.net kernel: SELinux:  policy capability cgroup_seclabel=1
Feb 07 13:41:27 server.bahis.net kernel: SELinux:  policy capability nnp_nosuid_transition=1
Feb 07 13:41:27 server.bahis.net kernel: SELinux:  policy capability genfs_seclabel_symlinks=1
Feb 07 13:41:27 server.bahis.net setsebool[8261]: The named_write_master_zones policy boolean was
changed to 1 by root
```

18: SELinux- named

DNS-

- dig ns.bahis.net
- NOERROR
- A- 192.168.1.1
- 127.0.0.1#53

```
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 54331
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x00b4, udp: 512
;; QUESTION SECTION:
;ns.user.net.                   IN      A

;; AUTHORITY SECTION:
user.net.               180     IN      SOA     ns0.dnsmadeeasy.com. dns.dnsmade
easy.com. 2006010168 43200 3600 1209600 180

;; Query time: 71 msec
;; SERVER: 172.9.0.14#53(172.9.0.14)
```

- host -l bahis.net
- NS- A-
-

```
[root@server.bahis.net rz]# host -t  A bahis.net
bahis.net has address 188.114.97.1
bahis.net has address 188.114.96.1
[root@server.bahis.net rz]# host -t PTR 192.168.1.1
Host 1.1.168.192.in-addr.arpa. not found: 3(NXDOMAIN)
[root@server.bahis.net rz]#
```

20: host -l bahis.net

- `host -a bahis.net`
- SOA, NS, A-
-

- A
- PTR
- 1.168.192.in-addr.arpa

2.5

- /vagrant/provision/server/dns
- named.conf
- provisioning

```
[root@server.bahis.net ~]# cd /var/named
[root@server.bahis.net named]# mkdir -p /var/named/master/fz
[root@server.bahis.net named]# mkdir -p /var/named/master/rz
[root@server.bahis.net named]# cp /var/named/named.localhost /var/named/master/f
z/
[root@server.bahis.net named]# cd /var/named/master/fz/
[root@server.bahis.net fz]# mv named.localhost bahis.net
[root@server.bahis.net fz]# ls
bahis.net
[root@server.bahis.net fz]# vim  /var/named/master/fz/bahis.net
```

21: DNS /vagrant

- dns.sh
- bind bind-utils
- /etc /var/named
- named:named
- SELinux-
- DNS firewall
- named_write_master_zones
- DNS 127.0.0.1
- named

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install bind bind-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/dns/etc/* /etc
cp -R /vagrant/provision/server/dns/var/named/* /var/named
chown -R named:named /etc/named
chown -R named:named /var/named
restorecon -vR /etc
restorecon -vR /var/named
echo "Configure firewall"
firewall-cmd --add-service=dns
firewall-cmd --add-service=dns --permanent
echo "Tuning SELinux"
```

- Vagrantfile provision- shell
- provision/server/dns.sh
-

```
## Server configuration
config.vm.define "server", autostart: fal
server.vm.box = "rocky9"
server.vm.hostname = 'server'
server.vm.boot_timeout = 1440
server.ssh.insert_key = false
server.ssh.username = 'vagrant'
server.ssh.password = 'vagrant'
server.vm.network :private_network,
ip: "192.168.1.1",
virtualbox__intnet: true
server.vm.provision "server dummy",
type: "shell",
preserve_order: true,
path: "provision/server/01-dummy.sh"
server.vm.provision "server dns"
```

3

3.

3.1

- DNS- BIND
-
- `bahis.net` `1.168.192.in-addr.arpa`
- A- PTR-
-
- DNS firewall
- SELinux
- DNS (127.0.0.1)
- `dns.sh` `Vagrantfile`